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1) Let $X \sim N(\mu, 1)$

we have 'n' i.i.d. realizations of X
while taking down the observations, we only noted whether (a) not value is +ve
Suppose m of n observations have -ve values
find the MLE of μ based on these observations.

Solution:

$$IP[X < 0] = IP\left[\frac{X - \mu}{1} < \frac{0 - \mu}{1}\right]$$

$$\begin{aligned} & \text{Standardize } X \\ &= IP[Z < -\mu] \quad Z \sim N(0, 1) \\ &= \phi(-\mu) \end{aligned}$$

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$$IP[X > 0] = 1 - IP[X < 0]$$

$$= 1 - \phi(-\mu)$$

$$= \phi(\mu)$$

using normal CDF
Inverse.

m : num of -ve observations.

$n-m$: num of +ve observations.

for each observation

$$IP[\text{Positive}] = \phi(\mu)$$

$$IP[\text{Negative}] = \phi(-\mu)$$

Now Likelihood of observed data

$$\begin{aligned} L(\mu) &= (IP[x < 0])^m (IP[x > 0])^{n-m} \\ &= (\phi(-\mu))^m (\phi(\mu))^{n-m} \end{aligned}$$

Now taking Log.

$$\ell(\mu) = \log L(\mu)$$

$$= \log \{ (\phi(-\mu))^m (\phi(\mu))^{n-m} \}$$

$$= m \log \phi(-\mu) + (n-m) \log \phi(\mu)$$

We can simplify by substituting

$$\begin{aligned} p &= IP[x < 0] \\ &= \phi(-\mu) \end{aligned}$$

$$\begin{aligned} 1-p &= IP[x > 0] \\ &= \phi(\mu) \end{aligned}$$

now the likelihood becomes

$$L(p) = p^m (1-p)^{n-m}$$

This is exactly the likelihood of Bernoulli model
-ve is treated as Success

m Success are observed in n-trials.

$$\ell(p) = m \log p + (n-m) \log (1-p)$$

differentiate wrt p

$$\frac{d}{dp} \ell(p) = \frac{m}{p} - \frac{n-m}{1-p}$$

equating to zero

$$\frac{m}{p} - \frac{n-m}{1-p} = 0$$

$$m(1-p) = (n-m)p$$

$$p = \frac{m}{n} \Rightarrow \hat{p} = \frac{m}{n}$$

convert $\hat{p}$ back to $\hat{\mu}$

$$p = \phi(-\hat{\mu})$$

$$\hat{p} = \phi(-\hat{\mu})$$

$$\frac{m}{n} = \phi(-\hat{\mu})$$

wk t

$$\phi(a) = b \Rightarrow a = \phi^{-1}(b)$$

$$a = -\hat{\mu}$$

$$b = \frac{m}{n}$$

$$-\hat{\mu} = \phi^{-1}\left(\frac{m}{n}\right) \Rightarrow \hat{\mu} = -\phi^{-1}\left(\frac{m}{n}\right)$$

$$\hat{\mu} = \phi^{-1}\left(1 - \frac{m}{n}\right)$$

wk t

$$-\phi^{-1}(q) = \phi^{-1}(1-q)$$

$$\boxed{\hat{\mu} = \phi^{-1}\left(\frac{n-m}{n}\right)}$$

2) Density of X is mixture of 2 exponential densities.

$$f(x; \theta) = \pi \beta_1 e^{-\beta_1 x} + (1-\pi) \beta_2 e^{-\beta_2 x}$$

$$\theta = \{ \pi, \beta_1, \beta_2 \} \quad 0 \leq \pi \leq 1, \beta_1 > 0, \beta_2 > 0$$

Use EM algorithm.

Use a scalar binary Z as a latent variable and show E step, Q, M step.

Solution:

$$Z = \begin{cases} 1 & \text{if } x \text{ comes from component 1} \\ 0 & \text{if } x \text{ comes from component 2} \end{cases}$$

$$Z \sim \text{Bernoulli}(\pi)$$

$x_1, x_2, \dots, x_n$ observations.

for each x_i , there is hidden variable z_i .

$\therefore$ Complete data is

$$(x_1, z_1), (x_2, z_2), \dots, (x_n, z_n)$$

$$f(x, z; \theta) = \prod_{i=1}^n f(x_i, z_i; \theta)$$

$$= \prod_{i=1}^n \left(\pi \beta_1 e^{-\beta_1 x_i} \right)^{z_i} \left((1-\pi) \beta_2 e^{-\beta_2 x_i} \right)^{(1-z_i)}$$

taking log,

$$l(\theta) = \log (f(x, z; \theta))$$

$$= \sum_{i=1}^n \left\{ \log \left(\pi \beta_1 e^{-\beta_1 x_i} \right)^{z_i} + \log \left((1-\pi) \beta_2 e^{-\beta_2 x_i} \right)^{(1-z_i)} \right\}$$

$$= \sum_{i=1}^n \left\{ z_i (\log \pi + \log \beta_1 - \beta_1 x_i) + (1-z_i) (\log (1-\pi) + \log \beta_2 - \beta_2 x_i) \right\}$$

In E step : Compute expected value of z_i , given observed data & current parameter estimate

$$\theta^{\text{old}} = \{ \pi^{\text{old}}, \beta_1^{\text{old}}, \beta_2^{\text{old}} \}$$

$$\gamma_i = \mathbb{E} [z_i | x_i ; \theta^{\text{old}}]$$

$$= \mathbb{P} [z_i = 1 | x_i ; \theta^{\text{old}}]$$

$$= \frac{\int (x_i | z_i = 1 ; \theta^{\text{old}}) \mathbb{P}[z_i = 1 ; \theta^{\text{old}}]}{\int (x_i ; \theta^{\text{old}})}$$

$$= \frac{\pi^{\text{old}} \beta_1^{\text{old}} e^{-\beta_1^{\text{old}} x_i}}{\pi^{\text{old}} \beta_1^{\text{old}} e^{-\beta_1^{\text{old}} x_i} + (1 - \pi^{\text{old}}) \beta_2^{\text{old}} e^{-\beta_2^{\text{old}} x_i}}$$

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$$\mathbb{P}[z_i = 0 | x_i ; \theta^{\text{old}}] = 1 - \gamma_i$$

$$\mathbb{E} [1 - z_i | x_i ; \theta^{\text{old}}] = 1 - \gamma_i$$

Now the Q-function is,

$$Q(\theta | \theta^{\text{old}}) = \mathbb{E}_{z|x;\theta^{\text{old}}} [\ell(\theta)]$$

$$= \mathbb{E}_{\mathbf{z}|\mathbf{x}; \theta^{old}} [\log p(\mathbf{x}, \mathbf{z}; \theta) | \mathbf{x}; \theta^{old}]$$

$$= \mathbb{E} \left[\sum_{i=1}^n \left\{ \overbrace{z_i (\log \pi + \log \beta_1 - \beta_1 x_i)}^{A_i} + \right. \right. \\ \left. \left. \overbrace{(1-z_i) (\log (1-\pi) + \log \beta_2 - \beta_2 x_i)}^{B_i} \right\} \right]$$

$$\begin{aligned} \mathbb{E}[z_i A_i] &= \mathbb{E}[z_i] A_i \\ &= \gamma_i A_i \end{aligned}$$

$$\begin{aligned} \mathbb{E}[(1-z_i) B_i] &= \mathbb{E}[1-z_i] B_i \\ &= (1-\gamma_i) B_i \end{aligned}$$

$$= \sum_{i=1}^n \left\{ \gamma_i (\log \pi + \log \beta_1 - \beta_1 x_i) + \right. \\ \left. (1-\gamma_i) (\log (1-\pi) + \log \beta_2 - \beta_2 x_i) \right\}$$

Let's do the M step.

$$Q_{\pi} = \sum_{i=1}^n \{ \gamma_i \log \pi + (1-\gamma_i) \log (1-\pi) \}$$

$$\frac{\partial}{\partial \pi} Q_{\pi} = \sum_{i=1}^n \left\{ \frac{\gamma_i}{\pi} - \frac{1-\gamma_i}{1-\pi} \right\}$$

equating to zero

$$\frac{1}{\pi} \sum_{i=1}^n \gamma_i = \frac{1}{1-\pi} \sum_{i=1}^n (1-\gamma_i)$$

$$S = \sum_{i=1}^n \gamma_i \Rightarrow \sum_{i=1}^n (1-\gamma_i) = n-S,$$

$$\frac{S}{\pi} = \frac{n-S}{1-\pi}$$

$$\pi^{\text{new}} = \frac{1}{n} \sum_{i=1}^n \gamma_i$$

M step for β_1

$$Q_{\beta_1} = \sum_{i=1}^n y_i (\log \beta_1 - \beta_1 x_i)$$

differentiate wrt β_1 & equate to zero.

$$\sum_{i=1}^n y_i \left(\frac{1}{\beta_1} - x_i \right) = 0$$

$$\frac{1}{\beta_1} \sum_{i=1}^n y_i = \sum_{i=1}^n y_i x_i$$

$$\beta_1^{\text{new}} = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n y_i x_i}$$

III) for β_2

$$\beta_2^{\text{new}} = \frac{\sum_{i=1}^n (1 - y_i)}{\sum_{i=1}^n (1 - y_i) x_i}$$

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